

SWIGGY SALES & CUSTOMER ANALYTICS REPORT

End-to-End Food Delivery Performance Analysis using Microsoft Fabric, SQL, Power BI & Power Query

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1. Executive Summary

This report presents a comprehensive analysis of Swiggy food delivery sales, customer ratings, restaurant performance, and regional order trends using Microsoft Fabric, SQL, Power BI, and Power Query.

The project was developed to evaluate food delivery business performance across multiple operational dimensions including sales growth, customer ordering behavior, food preferences, restaurant contribution, and regional performance trends.

The analysis combines data engineering, business intelligence, and visualization techniques to transform raw transactional food delivery data into actionable business insights.

A centralized Microsoft Fabric environment was used to integrate, clean, transform, and visualize the data. SQL was used for querying and KPI analysis, while Power Query was utilized for data cleaning and transformation. Interactive Power BI dashboards were developed to support business decision-making.

The dashboard reveals strong business growth with total sales exceeding ₹265M and nearly 1 million total orders processed.

Key Findings:

Metric	Value
Total Sales	₹265.06M
Average Order Value	₹268.51
Total Orders	987.15K
Average Rating	4.29M
Rating Count	28M

The analysis identified:

- Strong monthly revenue consistency
- High-performing restaurant chains contributing significantly to sales
- Karnataka as the highest-performing state
- Non-Veg category contributing the majority share of sales
- Weekend order spikes indicating peak demand periods

2. Business Context & Objectives

Business Context

Online food delivery platforms generate massive volumes of transactional and customer interaction data daily. Understanding customer preferences, regional demand patterns, restaurant performance, and revenue trends is critical for improving operational efficiency and customer satisfaction.

Swiggy operates across multiple cities and restaurant categories where business success depends heavily on:

- Customer retention
- Fast delivery operations
- Restaurant partnerships
- Order volume optimization
- Regional demand forecasting
- Pricing strategies

Project Objectives

The primary objectives of this analysis were:

- Analyze overall sales performance and order trends
- Identify high-performing restaurants and states
- Evaluate customer food preferences
- Measure average order value and customer ratings
- Track monthly, weekly, and daily sales trends
- Create an interactive business intelligence dashboard
- Build a centralized analytics solution using Microsoft Fabric
- Support data-driven business decision-making

3. Technology Stack

The project was developed using the following technologies:

Technology	Purpose
Microsoft Fabric	Data storage, analytics environment, and dashboard integration
SQL	Data querying, KPI calculations, and analysis
Power BI	Dashboard creation and visualization
Power Query	Data cleaning and transformation
DAX	KPI measures and calculations
Excel/CSV Dataset	Source transactional data

4. Data Collection & Architecture

Data Source

The dataset contains food delivery transaction data including:

- Order information
- Restaurant details
- Customer ratings
- Food category details
- State-wise sales
- Order values
- Time-based sales data

Microsoft Fabric Architecture

The project follows a modern analytics architecture using Microsoft Fabric.

Workflow:

Raw Dataset → Power Query Cleaning → SQL Analysis → Microsoft Fabric Warehouse → Power BI Dashboard

Data Pipeline Process

1. Imported raw food delivery dataset into Microsoft Fabric
 2. Cleaned and transformed data using Power Query
 3. Loaded transformed data into Fabric Warehouse
 4. Performed SQL-based KPI analysis
 5. Built Power BI dashboard connected to Fabric Warehouse
 6. Published interactive dashboard for reporting and business analysis
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5. Data Cleaning & Transformation

Data cleaning and preprocessing were performed using Power Query to ensure data quality, consistency, and analytical accuracy.

Cleaning Activities Performed

Missing Value Handling

- Removed null and duplicate records
- Replaced missing categorical values where necessary

Data Standardization

- Standardized restaurant names
- Corrected inconsistent food category values
- Formatted date and time columns

Data Type Conversion

- Converted sales columns into currency format
- Converted date fields into proper datetime format
- Corrected numerical data types for aggregation

Feature Engineering

Additional analytical columns were created including:

- Monthly Sales
- Weekly Sales
- Daily Sales
- Average Order Value
- Food Type Classification
- State-wise Performance Metrics

Benefits of Data Cleaning

Proper cleaning improved:

- Dashboard accuracy
 - Reporting consistency
 - KPI reliability
 - Query performance
 - Data modeling efficiency
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6. Data Modeling & SQL Analysis

SQL was used extensively for:

- KPI calculations
- Data aggregation
- Sales analysis
- Restaurant ranking
- Trend analysis
- State-wise performance evaluation

Example SQL Analysis Areas

Total Sales Calculation

- Sum of all completed order values

Average Order Value

- Average customer spending per order

Restaurant Ranking

- Top restaurants based on revenue contribution

State-wise Sales Analysis

- Revenue contribution by state

Trend Analysis

- Monthly and weekly sales tracking

Data Modeling

Relationships were created between:

- Orders table
- Restaurant table
- Food category table
- Time dimension table
- Regional dimension table

The star schema model improved dashboard performance and enabled efficient drill-down analysis.

7. Dashboard Overview

An interactive Power BI dashboard was developed to provide business stakeholders with real-time analytical insights.

Dashboard Features

KPI Cards

The dashboard includes:

- Total Sales
- Average Order Value
- Total Orders
- Average Ratings
- Rating Count

Filters & Slicers

Users can dynamically filter data by:

- City
- Food Type
- Quarter
- Restaurant Name

Visualizations Included

- Monthly Trend Analysis
 - Daily Sales Trend
 - Weekly Trend Analysis
 - Food Type Sales Distribution
 - Top Restaurant Sales Analysis
 - Top State Performance Analysis
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8. Key Performance Indicators (KPIs)

The following KPIs were calculated to evaluate business performance.

KPI	Value	Business Meaning
Total Sales	₹265.06M	Overall revenue generated
Average Order Value	₹268.51	Average customer spending
Total Orders	987.15K	Total orders processed
Average Rating	4.29	Customer satisfaction indicator
Rating Count	28M	Total customer engagement

KPI Insights

- Strong total sales indicate healthy business performance
 - High order volume reflects strong platform demand
 - Average order value suggests stable customer purchasing behavior
 - High rating score indicates positive customer experience
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9. Sales Performance Analysis

Monthly Sales Trend

The monthly sales trend shows relatively stable revenue generation across all months.

Key observations:

- Highest sales observed in May and August
- February recorded comparatively lower revenue
- Revenue remained consistently above ₹31M monthly

This consistency indicates strong customer demand and operational stability.

Daily Trend Analysis

Daily sales remained highly stable throughout the week.

Key findings:

- Weekend sales slightly outperform weekdays
- Saturday generated the highest sales volume
- Consistent daily revenue indicates predictable customer demand patterns

Weekly Trend Analysis

Weekly trends reveal:

- Stable order distribution across weeks
 - No major revenue fluctuations
 - Sustained business performance over time
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10. Customer & Rating Analysis

Customer satisfaction was evaluated using ratings and ordering behavior.

Rating Analysis

The platform achieved:

- Average rating of 4.29
- More than 28 million rating interactions

This reflects:

- Strong customer satisfaction
- High customer engagement
- Reliable service quality

Customer Ordering Behavior

Analysis suggests:

- Customers place orders consistently across weekdays and weekends
- Higher activity observed during peak dining periods
- Customers demonstrate stable purchasing behavior

11. Restaurant Performance Analysis

Restaurant-level analysis identified top-performing restaurant partners.

Top Restaurants by Sales

The highest-performing restaurants include:

- KFC
- McDonald's
- Pizza Hut
- Burger King
- Domino's Pizza

Key Insights

- International fast-food chains contribute significantly to overall revenue
- High-performing restaurants demonstrate strong brand loyalty
- Large restaurant chains generate higher transaction volumes

Business Importance

Restaurant performance analysis helps:

- Optimize partnership strategies
- Improve promotional planning
- Identify high-value restaurant collaborations

12. Regional Performance Analysis

State-wise analysis was conducted to identify geographical business performance.

Top Performing States

The highest revenue-generating states include:

- Karnataka
- Uttar Pradesh
- Telangana
- Maharashtra
- Delhi

Key Findings

- Karnataka generated the highest sales contribution
- Major metropolitan regions dominate revenue generation
- Urban population density strongly influences order volume

Business Impact

Regional analysis supports:

- Expansion planning
 - Regional marketing strategies
 - Delivery resource optimization
 - Localized promotional campaigns
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13. Trend Analysis

Food Type Analysis

Sales distribution by food category indicates:

- Non-Veg food contributes approximately 64.68% of sales
- Veg category contributes approximately 35.32%

Business Insights

- Non-Veg category dominates customer preference
- Food category trends can support menu optimization
- Marketing campaigns can be personalized based on food preferences

Time-Based Trends

The dashboard highlights:

- Consistent monthly performance
- Stable weekly demand
- Predictable customer ordering behavior

These trends enable:

- Better inventory management
 - Workforce planning
 - Delivery scheduling optimization
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14. Business Insights

The analysis generated several important business insights.

Major Insights

Strong Revenue Performance

The platform achieved over ₹265M in sales, indicating strong market demand and customer retention.

High Customer Engagement

The platform recorded nearly 1 million orders and 28 million ratings, reflecting active customer participation.

Non-Veg Category Dominance

Non-Veg food contributes the majority share of total sales, making it a key revenue driver.

Restaurant Chain Dominance

Large fast-food chains contribute significantly to platform revenue due to strong customer demand and brand recognition.

Regional Revenue Concentration

Karnataka and major metropolitan states generate the highest order volume and sales.

Stable Demand Patterns

Consistent monthly and weekly trends indicate predictable business performance.

15. Strategic Recommendations

Based on the analysis, the following business recommendations are proposed.

HIGH PRIORITY

1. Strengthen High-Performing Restaurant Partnerships

Collaborate closely with top-performing restaurant chains to increase platform revenue and customer retention.

2. Expand Non-Veg Category Promotions

Since Non-Veg orders dominate revenue contribution, targeted marketing campaigns should focus on this category.

3. Focus on High-Revenue States

Increase marketing investments and delivery infrastructure in Karnataka and other high-performing states.

MEDIUM PRIORITY

4. Improve Customer Personalization

Use customer ordering patterns and food preferences for personalized recommendations.

5. Optimize Weekend Delivery Operations

Allocate additional delivery resources during weekends to manage higher order volumes efficiently.

6. Develop Loyalty Programs

Introduce customer rewards and loyalty programs to increase repeat purchases.

LOW PRIORITY

7. Enhance Rating Analysis

Perform deeper sentiment analysis on customer ratings and feedback to improve service quality.

8. Forecast Future Demand

Implement predictive analytics models for demand forecasting and operational planning.

16. Conclusion

This project successfully demonstrates the use of Microsoft Fabric, SQL, Power BI, and Power Query to build an end-to-end food delivery analytics solution.

The dashboard transformed raw food delivery transaction data into meaningful business intelligence capable of supporting strategic decision-making.

The analysis identified:

- Strong sales performance

- High customer engagement
- Dominance of Non-Veg food categories
- High-performing restaurant chains
- Regional revenue concentration in major states
- Stable business growth trends

The project also highlights the importance of modern data analytics platforms such as Microsoft Fabric in building scalable, centralized, and efficient business intelligence ecosystems.

By integrating data engineering, SQL analytics, data transformation, and interactive visualization, this project delivers a complete analytics solution for food delivery business operations.

Tools & Skills Demonstrated

- Microsoft Fabric
 - SQL
 - Power BI
 - Power Query
 - Data Cleaning
 - Data Transformation
 - Data Modeling
 - Dashboard Development
 - KPI Analysis
 - Business Intelligence
 - Data Visualization
 - Analytical Reporting
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Project Title

Swiggy Sales & Customer Analytics using Microsoft Fabric | SQL | Power BI | Power Query